

The marginal propensity to consume in heterogeneous agent models

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Introduction

1. **Motivation.** MPC a key object in het-agent incomplete-market models w/ nominal rigidities. It determines

- the size of fiscal multipliers (Kaplan & Violante 2014, Auclert et al. 2018)
- transmission of monetary policy (Kaplan et al. 2018, Auclert 2019)
- amplification of aggregate shocks (Werning 2015)

⇒ Need models that can reproduce the data on MPC

2. **Definition of MPC:** How much a household spends, within a given period, out of a small, one-time, unanticipated windfall.

Focus on

- Period: **One quarter**
- Amount: **\$500**
- Spending: **nondurable goods and services**
- MPC out of **transitory** shock
- **Contemporaneous** MPC, (MPC on impact, not e.g. MPC out of news)

This paper

1. Empirical evidence: Quarterly aggregate MPC 15%-25%.
 - Substantial heterogeneity: Many zeros, many close to one
 - Sign and size asymmetry: smaller windfall $>$ larger windfall, negative windfall $>$ positive windfall
2. Question: How close can HA models come to match the evidence on MPC, when constrained to match key moments from wealth distributions?
3. Approach: Solve multiple versions the model, under different calibration strategies.

Findings (one-asset model)

1. Matching aggregate wealth, **MPC 3%-5%**. $\gg$ RA models, $\ll$ data
2. Matching liquid wealth, **MPC $\geq$ data**, but abstract from 98% of aggregate wealth
3. Adding ex-ante heterogeneity or behavioral preferences, **MPC & aggregate wealth can match data**, but has the **missing middle** problem, median wealth 5-10 times smaller than data
4. Spender-saver models, when carefully calibrated, **can reconcile these tensions**, but has implausible implication for time paths of MPC.

Findings (two-asset model)

1. Can match MPC and wealth data, no missing middle problem, but baseline requires a **sizable return gap (8%)** in liquid and illiquid assets
2. Extensions of two-asset model can achieve the above **w/ a smaller return gap**. Example:
 - Direct utility flow: owner-occupied housing
 - Indirect utility flow: preferences that feature temptation and self control

ONE-ASSET MODEL

What determines the size of the aggregate MPC

Shape of the consumption function (Caroll 2001)

1. Uninsurable risk $\implies$ precautionary saving motive
 - Prudence $u''' > 0$
 - Occasionally binding liquidity constraints
2. Strength of precautionary saving is decreasing in wealth
3. These imply consumption function concave in wealth $\implies$ MPC decreasing in wealth
4. As wealth grows, the MPC asymptotes to certainty equivalent case

Shape of the wealth distribution

1. Bigger mass at bottom, where consumption function is concave $\implies$ larger MPC
2. Hand-to-mouth (HtM) households with zero wealth/binding BC, have $MPC \approx 1$

Wealth distribution

1. Data: SCF 2019. Sample: Drop the top 5% of the wealth distribution
2. Mean annual earnings is \$67,000. Wealth expressed as multiples of mean annual earnings
3. Key moments from data:
 - Net worth: Mean is 4.1 (\$275,000). Median is 1.5 (\$100,500)
 - Liquid wealth: Mean is 0.56 (\$37,500). Median is 0.05 (\$3,100) (Extremely right-skewed)
4. Definition of HtM:
 - One asset: $\text{Wealth} \leq \frac{1}{2} \times \text{monthly income}$. Data: 14% HtM .
 - Two asset: $\text{Liquid wealth} \leq \frac{1}{2} \times \text{monthly income}$. Data: 41% HtM. 27% wealthy, 14% poor

Baseline one-asset model

1. Household's income fluctuation problem: $(\beta = \tilde{\beta}(1 - \delta))$

$$\max_{\{c_t, b_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}$$

$$\text{s.t. } c_t + b_{t+1} = Rb_t + y_t, \quad b_{t+1} \geq 0, \quad b_0 = 0$$

$$\beta R < 1 \implies \text{solution: } \{c(b, y), b'(b, y), \mu(b, y)\}$$

2. Parametrization (quarterly):

- Partial equilibrium: $r = 0.25\%$
- Log utility: $\gamma = 1$
- Process for y_t : persistent + transitory components (PSID)
- $\delta = \frac{1}{200}$: match an adult life span of 50 years
- β set to match aggregate wealth-income ratio of 4.1 (SCF)

3. Without income uncertainty and borrowing constraint, MPC $m_0^* = 1 - R^{-1}(\beta R)^{\frac{1}{\gamma}}$.

$$\gamma = 1 \implies m_0^* = 1 - \beta. \text{ (PIH also has this) Baseline calibration implies } m_0^* = 0.5\%$$

MPC calculation

1. **Individual MPC** for HH with state (b, y) is calculated as

$$m_0(x; b, y) = \frac{c(b+x, y) - c(b, y)}{x}$$

2. **Aggregate MPC**

$$\bar{m}_0 = \int_{B \times Y} m_0(x; b, y) d\mu(b, y)$$

3. **Key determinants for aggregate MPC:**

- **Consumption function $c(b, y)$** $\implies$ MPC function $m_0(b, y)$
- **Stationary distribution $\mu(b, y)$**

Baseline result

	Data	Baseline
Quarterly MPC (%)	NA	4.6
Share of HtM (%)	14.2	2.5
Annualized discount factor	NA	0.980
<i>Panel B: Wealth statistics</i>		
Mean wealth	4.1	4.1
Median wealth	1.5	1.3
$a \leq \$1,000$	15.1	2.5
$a \leq \$5,000$	19.5	11.6
$a \leq \$10,000$	24.6	18.5
$a \leq \$15,000$	37.8	40.3
$a \leq \$100,000$	49.4	51.9
Wealth, top 10% share	49.9	46.6

Table: Baseline one-asset model

MPC function and wealth distribution

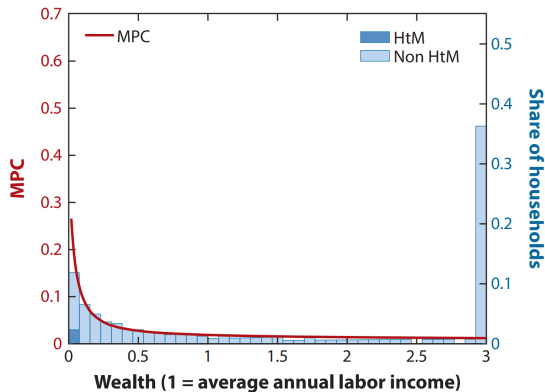


Figure: Baseline one asset: MPC function (left scale) and HH wealth distribution (right scale)

1. What if we target other wealth levels?

Sensitivity – A useful decomposition

1. Can decompose changes in the aggregate MPC into three components
2. Let $\mathbf{m}_0^*(b, y)$, $\mu^*(b, y)$ be the baseline MPC function and the stationary distribution
3. The aggregate MPC in an alternative model can be written as

$$\begin{aligned}\bar{\mathbf{m}}_0 = & \underbrace{\bar{\mathbf{m}}_0^*}_{\text{benchmark MPC}} + \underbrace{\int_{\mathcal{B} \times \mathcal{Y}} [\mathbf{m}_0 - \mathbf{m}_0^*](b, y) d\mu^*(b, y)}_{\text{consumption function}} + \underbrace{\int_{\mathcal{B} \times \mathcal{Y}} \mathbf{m}_0^*(b, y) d[\mu - \mu^*](b, y)}_{\text{distribution}} \\ & + \underbrace{\int_{\mathcal{B} \times \mathcal{Y}} [\mathbf{m}_0 - \mathbf{m}_0^*](b, y) d[\mu - \mu^*](b, y)}_{\text{interaction}}\end{aligned}$$

Sensitivity under different calibrations

	Baseline	top 5% included		target liquid wealth		
		Mean(a)	Median(a)	Mean(a)	Median(a)	HtM
Quarterly MPC (%)	4.6	2.7	4.3	14.0	33.7	22.0
Share of HtM (%)	2.5	1.7	2.3	7.5	37.4	14.2
Annualized discount factor	0.980	0.988	0.981	0.945	0.826	0.909
<i>Panel A: Decomposition</i>						
Gap w/ baseline MPC	NA	-1.9	-0.3	9.4	29.1	17.4
Effect of MPC	NA	-0.8	-0.1	3.1	11.1	6.0
Effect of distribution	NA	-1.3	-0.2	4.6	12.5	7.7
Interaction	NA	0.2	0.0	1.7	5.6	3.7
<i>Panel B: Wealth statistics</i>						
Mean wealth	4.1	9.4	4.6	0.6	0.1	0.3
Median wealth	1.3	3.5	1.5	0.2	0.05	0.1
Wealth, top 10% share	46.6	44.6	46.1	52.0	56.5	51.6

1. What if we change the parameter values?

Sensitivity under different parameter values

	Baseline	$r = 0\%$	$r = 5\%$	$\gamma = 0.5$	$\gamma = 5$
Quarterly MPC (%)	4.6	4.4	5.2	5.3	3.0
Share of HtM (%)	2.5	2.4	2.5	3.3	0.6
Annualized discount factor	0.980	0.990	0.944	0.986	0.840
<i>Panel A: Decomposition</i>					
Gap w/ baseline MPC	NA	-0.2	0.6	0.7	-1.6
Effect of MPC	NA	-0.2	0.7	0.1	0.5
Effect of distribution	NA	0.0	-0.1	0.6	-2.1
Interaction	NA	0.0	0.0	0.0	0.1
<i>Panel B: Wealth statistics</i>					
Mean wealth	4.1	4.1	4.1	4.1	4.1
Median wealth	1.3	1.4	1.4	1.1	2.3
Wealth, top 10% share	46.6	46.2	45.9	48.6	36.3

Take stock

1. There is a tension between data and model
2. Data has a lot of wealth and a high aggregate MPC
3. If the model targets aggregate wealth, then we need a high β
 - MPC at all wealth level decreases
 - HH save themselves away from the concave region of consumption function

HtM share low, MPC low
4. To generate a higher MPC, the model has to target a lower wealth $\implies$ a low β
 - MPC at all wealth level increases
 - More HH with low wealth around the concave region

HtM share high, MPC high, but aggregate wealth very small
5. Adding ex-ante heterogeneity (types) seems like a fix

Adding ex-ante heterogeneity

Consider a uniform distribution over grids

1. For heterogeneity in β , consider $\{\beta - 2\Delta, \beta - \Delta, \beta, \beta + \Delta, \beta + 2\Delta\}$ for $\Delta \in \{0.02, 0.04\}$
Choose mid point β to match aggregate wealth
2. For heterogeneity in r , consider $\{-1\%, 1\%, 3\%\}$, $\{-3\%, 1\%, 5\%\}$

Adding ex-ante heterogeneity - Results

	Baseline	Het β		Het r	
		$\Delta = 0.02$	$\Delta = 0.04$	$r \in \{-1, 1, 3\}$	$r \in \{-3, 1, 5\}$
Quarterly MPC (%)	4.6	11.7	18.6	7.5	11.6
Share of HtM (%)	2.5	6.7	14.2	4.0	6.8
Median annualized β	0.980	0.953	0.915	0.970	0.954
<i>Panel A: Decomposition</i>					
Gap w/ baseline MPC	NA	7.1	14.0	2.9	7.0
Effect of MPC	NA	2.4	4.3	1.2	2.7
Effect of distribution	NA	3.2	6.7	1.4	3.1
Interaction	NA	1.4	3.7	0.3	1.2
<i>Panel B: Wealth statistics</i>					
Mean wealth	4.1	4.1	4.1	4.1	4.1
Median wealth	1.3	0.3	0.2	0.6	0.3
Wealth, top 10% share	46.6	79.2	87.9	64.7	74.0

Table: Heterogeneity in one-asset model

MPC function and wealth distribution w/ het β

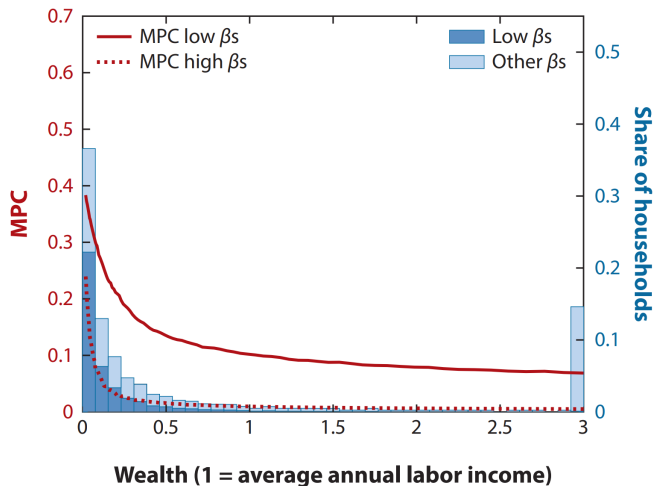


Figure: Heterogeneous β one-asset model

More heterogeneity

1. (Appendix) Heterogeneous γ : Similar story to het β
2. Epstein-Zin preference:

$$U_t = \left\{ (1 - \beta) c_t^{1-\theta} + \beta \left(\mathbb{E}_t \left[U_{t+1}^{\frac{1-\theta}{1-\gamma}} \right] \right)^{1-\gamma} \right\}$$

γ the RRA, $1/\theta$ the IES.

- Without heterogeneity, varying either does not raise aggregate MPC.
- With het γ holding IES fixed, barely any effect on aggregate MPC.
- With het IES holding γ fixed, can generate high aggregate MPC – similar story to het β

Behavioral preferences

1. Two alternative ways are proposed in the literature to generate high MPC.
 - Temptation and self control
 - Present bias
2. **Punchline**: Both are effectively models of **endogenous discount rates** that depend on asset level, hence have the same pros and cons as models w/ het β .

Temptation and self control

1. Household problem is

$$\begin{aligned} v(b, y) &= \max_{b'} \{ u(c) + \beta \mathbb{E}(v(b', y')) + \varphi(u(c) - u(Rb + y)) \} \\ \text{s.t.} \quad &c + b' = Rb + y, \quad b' \geq 0 \end{aligned}$$

$\varphi > 0$ measures the strength of temptation.

2. Euler equation is

$$u_c(c) = \beta R \mathbb{E} \left[\left(1 - \frac{\varphi}{1 + \varphi} \frac{u_c(Rb' + y')}{u_c(c')} \right) u_c(c') \right]$$

With log utility, the **effective discount factor** is ($\times$ the cash on hand)

$$\beta \mathbb{E} \left[\left(1 - \frac{\varphi}{1 + \varphi} \frac{c'}{x'} \right) \right]$$

the effective discount factor decreasing in APC. Empirical evidence: (Lewis, Melcangi & Pilossoph 2022)

Temptation – Results

	Baseline	Temptation		Het temptation
		$\varphi = 0.01$	$\varphi = 0.05$	$\varphi \in \{0, 0.05, 0.1\}$
Quarterly MPC (%)	4.6	6.4	19.1	21.4
Share of HtM (%)	2.5	4.0	20.4	29.3
<i>Panel A: Decomposition</i>				
Gap w/ baseline MPC	NA	1.8	14.5	16.8
Effect of MPC	NA	0.4	2.7	3.9
Effect of distribution	NA	1.2	7.4	8.3
Interaction	NA	0.2	4.4	4.5
<i>Panel B: Wealth statistics</i>				
Mean wealth	4.1	4.1	4.1	4.1
Median wealth	1.3	1.0	0.1	0.1
Wealth, top 10% share	46.6	49.8	70.6	70.1

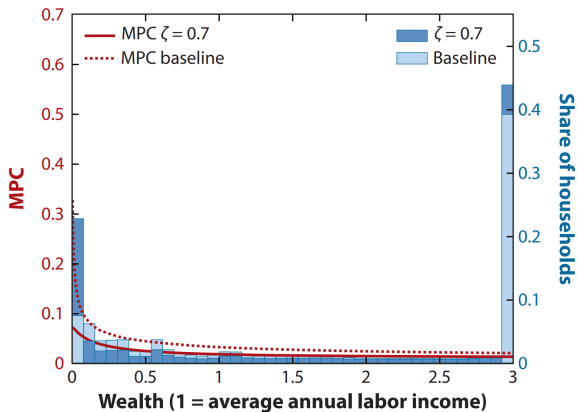
Present bias

1. Time **inconsistent** preferences w/ **naive consumers**. **Instantaneous gratification**
2. Formulated in continuous time.
3. Key difference is the discounting function (Laibson et al. 2021)

$$D(t) = \begin{cases} 1 & t = 0 \\ \zeta e^{-\rho t} & t > 0 \end{cases}$$

where $\zeta < 1$ measures the present bias

MPC function & wealth distribution w/ present-bias



Aggregate MPC remains small

Two offsetting forces: 1. more low-wealth HH 2. higher calibrated discount factor lowers MPC

Spender-saver models

1. Parametrization: 15% spenders w/ $\beta^{spend} = 0.4$, 85% savers whose discount factor set to match aggregate wealth

	Baseline	Spender-saver
Quarterly MPC (%)	4.6	16.8
Share of HtM (%)	2.5	14.8
Median annualized β	0.980	0.392
<i>Panel A: Decomposition</i>		
Gap w/ baseline MPC	NA	12.2
Effect of MPC	NA	1.9
Effect of distribution	NA	3.3
Interaction	NA	7.0
<i>Panel B: Wealth statistics</i>		
Mean wealth	4.1	4.1
Median wealth	1.3	1.0
Wealth, top 10% share	46.6	50.9

TWO-ASSET MODEL

Baseline two-asset

1. Motivation for modeling two assets:
 - Not all household wealth **immediately available** for consumption smoothing
 - In line with evidence that **MPC is decreasing in liquid wealth**
2. Two assets: **illiquid (a)** and **liquid (b)** with $r^a > r^b$
3. Fixed transaction cost to move between liquid & illiquid accounts
4. Different precautionary and smoothing motives
 - Liquid asset: against **small, regular** income fluctuations. $\beta R^b \ll 1$ prevents HH from accumulating too much liquid wealth
 - Illiquid asset: against **large, irregular** income fluctuations.

Why wealthy HtM emerge

1. Three types of HH emerge in this economy
 - Unconstrained (59%)
 - **Poor HtM**: zero net worth (14%)
 - **Wealthy HtM**: zero liquid wealth, but positive illiquid wealth (27%)
2. Why holding zero liquidity and some illiquid wealth at the same time? A trade-off:
 - **Long-run gain**: higher level consumption from higher return (**first order**)
 - **Short-run cost**: worse consumption smoothing (**second order**)

Baseline two-asset – HH's problem

1. Continuous time. Household's HJB equation

$$\rho v(a, b, y) = \max_c \{ u(c) + \chi [v^*(a, b, y) - v(a, b, y)] + v_b(a, b, y) \dot{b} + v_a(a, b, y) \dot{a} + \mathcal{A}v(a, b, y) \}$$

subject to

$$\dot{b} = r^b b + y - c, \quad \dot{a} = r^a a, \quad a \geq 0$$

χ : Arrival rate of rebalancing opportunities

$\mathcal{A}$: Infinitesimal generator of the stochastic process for income

2. The value function after the rebalancing opportunity arises is given by

$$v^*(a, b, y) = \max \{ \omega(a + b, y), v(a, b, y) \}$$

where

$$\omega(a + b, y) = \max_{a', b'} v(a', b', y)$$

subject to

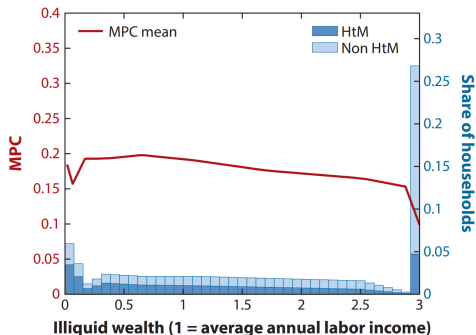
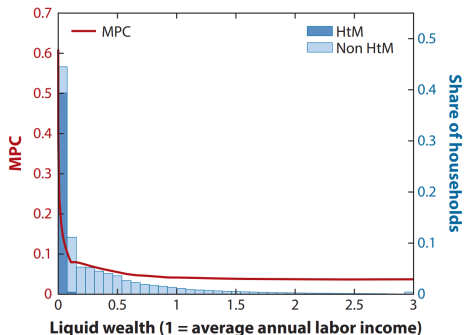
$$a' + b' \leq a + b - \kappa, \quad a', b' \geq 0$$

Baseline two-asset – Parametrization & result

Parameter	Value	Moment	Value
Discount factor ρ (annualized)	0.937	Mean wealth	4.1
Illiquid rate r^a	6.2 %	HtM share	41%
Transaction cost κ	\$1456.8	Poor HtM share	14%
Liquid rate r^b	-2%	Zero nominal return	
Rebalancing arrival rate χ	3	Once every month	

	data	Baseline one-asset (cts time)	Baseline two-asset
Quarterly MPC (%)	NA	3.0	16.1
Annualized discount factor (%)		0.985	0.937
Median total wealth	1.5	1.6	1.1
Mean liquid wealth	0.56	NA	0.2
Median liquid wealth	0.05	NA	0.1
$w \leq \$1,000$	15.1	2.5	10.0
$w \leq \$5,000$	19.5	8.0	20.8
$w \leq \$10,000$	24.6	13.4	23.5
$w \leq \$50,000$	37.8	34.9	45.9
$w \leq \$100,000$	49.4	48.0	57.7
Wealth, top 10% share	49.9	40.3	62.0

MPC function and liquid/illiquid wealth



1. What matters for the size of MPC is **liquid wealth**, not illiquid wealth
2. MPC remains high also for HH with sizable illiquid wealth (they hold little liquid wealth)

What about the return gap

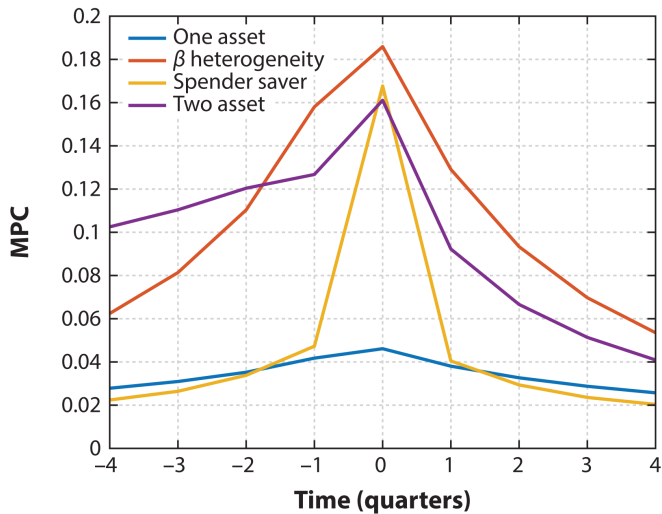
1. A significant impact of the return gap on the aggregate MPC: Gap $\uparrow$, more HtM, higher MPC, vice versa
2. Jordà et al. (2019): A gap of 5-6% between the return on stocks and housing versus treasury bills
 - KV: Did not take tax advantages into account
3. Owner-occupied housing $\approx$ a 3-4% financial return per year (Kaplan & Violante 2014)
4. Add temptation and self-control
 - May be difficult to save due to short-term gratification
 - Adds a commitment value, HH want to reduce temptation by locking wealth away in illiquid form

Two-asset w/ temptation and self-control

	Baseline two-asset	$(r^a - r^b)$ & κ fixed		$(r^a - r^b)$ & κ recalibrated	
		$\varphi = 0.01$	$\varphi = 0.05$	$\varphi = 0.01$	$\varphi = 0.015$
Quarterly MPC (%)	16.1	23.8	46.7	18.1	21.1
Share HtM	39.5	48.6	62.9	39.0	40.4
Share PHtM	13.2	18.0	24.0	13.0	13.3
Annualized liquid return	-2.0	-2.0	-2.0	-2.0	-2.0
Annualized illiquid return	6.2	6.2	6.2	3.2	3.2
Rebalance cost (\$)	1456.8	1456.8	1456.8	2400.0	2400.0
Mean total wealth	4.1	4.1	4.1	4.1	4.1
Median total wealth	1.1	1.0	0.7	1.0	1.0
Wealth, top 10% share	62.0	63.5	65.8	60.9	60.6

OTHER EVIDENCE

Intertemporal MPC



MPC size and sign asymmetry

	Baseline one asset	mean(liquid)	β Het	IES Het	S-S	Baseline two-asset
MPC (%) out of $-\$500$	5.0	14.9	19.5	22.7	17.2	26.6
MPC (%) out of $+\$500$	4.6	14.0	18.6	20.8	16.8	16.1
MPC (%) out of $+\$5,000$	3.6	11.3	15.3	14.5	13.2	12.3

1. All models display correct size and sign dependence: **concavity**

MPC out of illiquid wealth

1. Empirical counterpart: MPC out of changes in housing wealth, unrealized capital gains/losses on equity in retirement accounts or private business
2. Literature: Average MPC out of housing wealth:
 - Carroll et al. (2011): 2%
 - Mian et al. (2013): 1.5%, a lot of heterogeneity in income and leverage
3. Model: A small injection of illiquid wealth
 - +\$500: 1.4%
 - +\$5000: 2.4%

Conclusion

Can macro models generate a high MPC, and how?

1. Representative agent model: **NO!**
2. Canonical one-asset model: **only by neglecting 90% of wealth**
3. Ex-ante heterogeneity & behavioral preferences: **impoverish middle problem**
4. Spender-saver model: **implausible spending implication**
5. Two-asset model: **successful w/ a large enough return gap, a few justifications from theory and data**

THANKS!

APPENDIX

Appendix – HJB equation

1. Optimal control problem

$$v(x_0) = \max_{\{a(t)\}_{t \geq 0}} \int_0^{\infty} e^{-\rho t} r(x(t), a(t)) dt$$

subject to

$$\dot{x}(t) = g(x(t), a(t))$$

2. **Result:** The value function satisfies the Hamilton-Jacobi-Bellman equation

$$\rho v(x) = \max_a \{r(x, a) + v'(x) \cdot g(x, a)\}$$

3. The infinitesimal generator $\mathcal{A}$ can be thought of as the derivative w.r.t the stochastic process y . It is defined as

$$\mathcal{A}v(a, b, y) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}_t[v(a, b, y_{t+\Delta t}) | y_t = y] - v(a, b, y)}{\Delta t}$$

Appendix – Income process (Data)

1. Data: PSID 1968-2008. Drop income less than \$7,250 2016 dollars
2. Calculate 2nd and 4th moment of log earning (residualized) growth at different lags, $j = 2, 4$, and the kurtosis.

$$m_{j,d} = \begin{cases} \mathbb{E} \left[(\log y_t^{ann})^j \right] & d = 0 \\ \mathbb{E} \left[(\log y_{t+d}^{ann} - \log y_t^{ann})^j \right] & d > 1 \end{cases} \implies \kappa_d = \frac{m_{4,d}}{(m_{2,d})^2}$$

Lag (d)	0	1	2	3	4	5	6	7	8	9	10
$m_{2,d}$	0.504	0.142	0.207	0.235	0.280	0.295	0.335	0.352	0.383	0.398	0.422
$m_{4,d}$	0.930	0.220	0.369	0.410	0.544	0.557	0.694	0.729	0.838	0.885	0.967
κ_d	3.65	10.90	8.57	7.42	6.96	6.49	6.19	5.87	5.72	5.59	5.43

Appendix – Income process (Model)

1. The discrete-time income process

$$\log y_t = \begin{cases} z_t + \varepsilon_t & \text{w.p. } \lambda_\varepsilon, \quad \varepsilon \sim \mathcal{N}\left(-\frac{\sigma_\varepsilon^2}{2}, \sigma_\varepsilon^2\right) \\ z_t & \text{w.p. } 1 - \lambda_\varepsilon \end{cases}$$

where the persistent component follows

$$z_t = \begin{cases} \phi_z z_{t-1} + \eta_t & \text{w.p. } \lambda_\eta, \quad \eta_t \sim \mathcal{N}\left(-\frac{\sigma_\eta^2}{2}, \sigma_\eta^2\right) \\ \phi_z z_{t-1} & \text{w.p. } 1 - \lambda_\eta \end{cases}$$

2. Baseline calibration: Fix $\lambda_\eta, \lambda_\varepsilon = 0.25$, choose $\{\phi_z, \sigma_\eta^2, \sigma_\varepsilon^2\}$ to match $\{m_{2,0}, m_{2,1}, m_{2,5}\}$